Midterm Report: 5G Network Implementation using srsRAN Project and Open5GS

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1 Introduction

This report presents the midterm progress of implementing a 5G network using srsRAN Project and Open5GS. The srsRAN Project provides an open-source 5G RAN (Radio Access Network) implementation, while Open5GS serves as the 5G Core Network (5GC).

The goal of this project is to set up a fully functional 5G network on a server environment, analyze its performance, and document the configuration and results obtained.

1.1 Objectives

- Install and configure Open5GS as the 5G core network
- Install and configure srsRAN Project as the 5G RAN
- Establish end-to-end connectivity between RAN and Core
- Analyze and document network performance

2 Methodology

2.1 Server Environment

The implementation was carried out on a remote SSH server running Ubuntu. Both srsRAN Project and Open5GS were installed and configured on this server.

2.2 Open5GS Setup

Open5GS was installed following the official documentation. The core network components including AMF, SMF, UPF, and other network functions were configured to support 5G SA (Standalone) operation.

2.3 srsRAN Project Setup

The srsRAN Project gNB (next generation NodeB) was configured to connect to the Open5GS core. Configuration files were modified to set the correct PLMN, AMF address, and radio parameters.

2.4 Network Architecture

Figure 1 shows the 5G network architecture implemented using srsRAN Project and Open5GS.

“‘

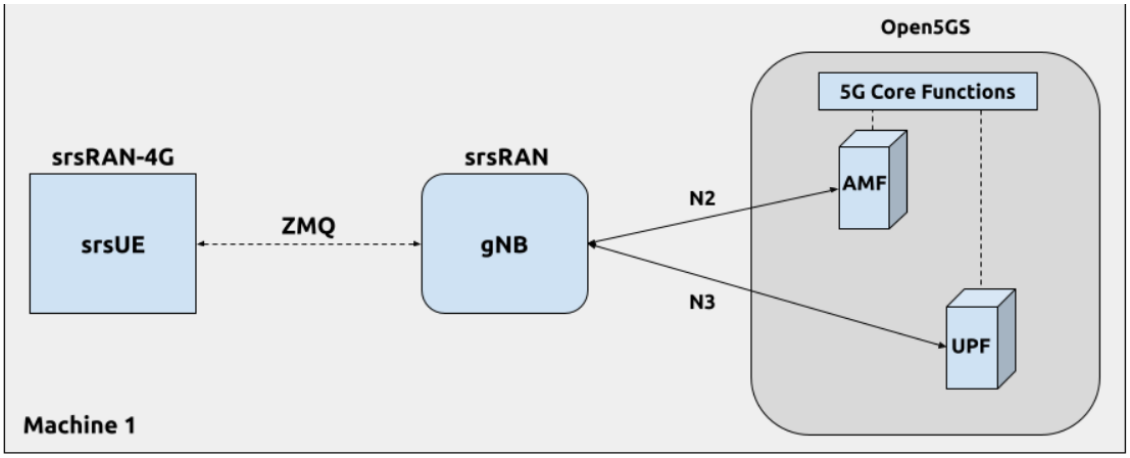


Figure 1: 5G Network Architecture with srsRAN and Open5GS

3 Results

3.1 Installation Status

Both srsRAN Project and Open5GS were successfully installed on the SSH server. The required dependencies and configurations were completed as part of the midterm milestone.

3.2 UE Connection to gNB

Figure 2 shows the UE successfully connected to the gNB.

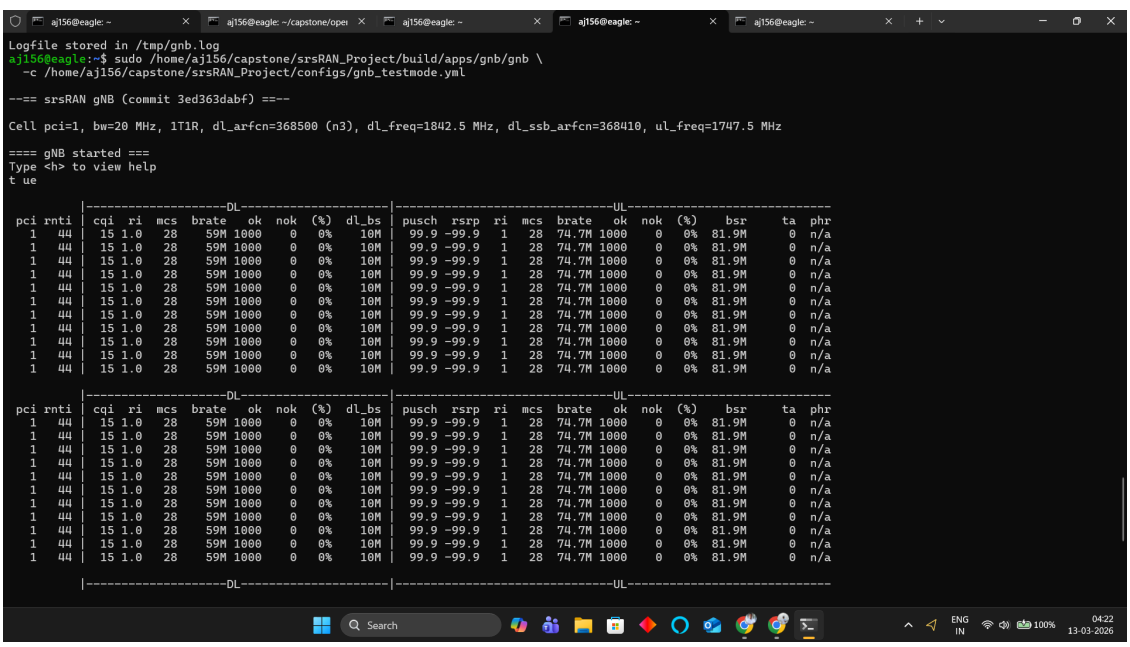
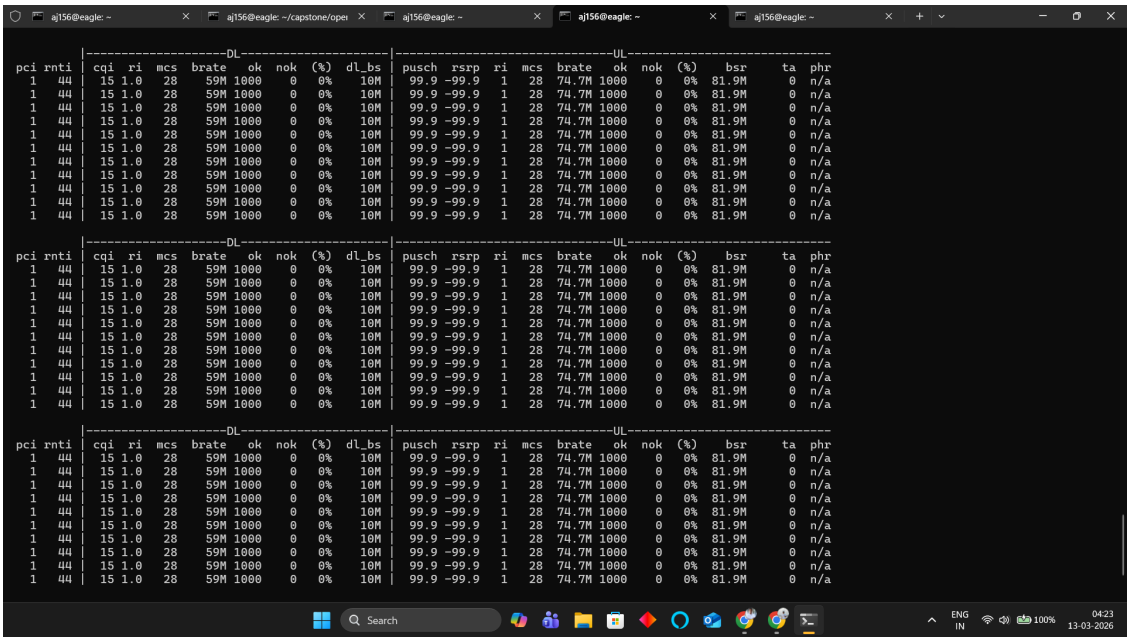


Figure 2: UE Successfully Connected to gNB

3.3 Uplink and Downlink Data

Figure 3 shows the uplink and downlink data transmission.



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aj156@eagle: ~   aj156@eagle: ~/capstone/open   aj156@eagle: ~   aj156@eagle: ~   aj156@eagle: ~

          |-------------------DL--------------------|-------------------------------UL-------------------------------
pci rnti | cqi  ri  mcs  brate   ok  nok  (%)  dl_bs | pusch  rsrp  ri  mcs  brate   ok  nok  (%)    bsr     ta  phr
  1   44 |  15 1.0   28    59M 1000    0   0%    10M |  99.9 -99.9   1   28  74.7M 1000    0   0%  81.9M      0  n/a
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          |-------------------DL--------------------|-------------------------------UL-------------------------------
pci rnti | cqi  ri  mcs  brate   ok  nok  (%)  dl_bs | pusch  rsrp  ri  mcs  brate   ok  nok  (%)    bsr     ta  phr
  1   44 |  15 1.0   28    59M 1000    0   0%    10M |  99.9 -99.9   1   28  74.7M 1000    0   0%  81.9M      0  n/a
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          |-------------------DL--------------------|-------------------------------UL-------------------------------
pci rnti | cqi  ri  mcs  brate   ok  nok  (%)  dl_bs | pusch  rsrp  ri  mcs  brate   ok  nok  (%)    bsr     ta  phr
  1   44 |  15 1.0   28    59M 1000    0   0%    10M |  99.9 -99.9   1   28  74.7M 1000    0   0%  81.9M      0  n/a
  1   44 |  15 1.0   28    59M 1000    0   0%    10M |  99.9 -99.9   1   28  74.7M 1000    0   0%  81.9M      0  n/a
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  1   44 |  15 1.0   28    59M 1000    0   0%    10M |  99.9 -99.9   1   28  74.7M 1000    0   0%  81.9M      0  n/a
  1   44 |  15 1.0   28    59M 1000    0   0%    10M |  99.9 -99.9   1   28  74.7M 1000    0   0%  81.9M      0  n/a
  1   44 |  15 1.0   28    59M 1000    0   0%    10M |  99.9 -99.9   1   28  74.7M 1000    0   0%  81.9M      0  n/a

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Figure 3: Uplink and Downlink Data Transmission

3.4 Configuration

The following key configurations were applied:

- Open5GS AMF configured with PLMN and slice information
- srsRAN gNB connected to Open5GS AMF via NGAP interface
- UE registration and authentication tested

3.5 Current Progress

Open5GS installation complete

srsRAN Project installation complete

Basic configuration completed

UE connected to gNB

Full end-to-end testing in progress

3.6 Next Steps

The remaining work includes completing end-to-end UE connectivity testing, performance benchmarking, and analysis of the results using MATLAB. “‘

References

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